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Synthesize and characterization of graphene nanosheets with high surface area and nano-porous structure

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ABSTRACT

A few-layer graphene was obtained by the expansion and exfoliation of water-intercalated graphene oxide via heat treatment in nitrogen environment in the temperature range of 200–1000 °C. Graphene which was synthesized at 800 °C (GT800) had a higher quality than other temperatures. This graphene has a high specific surface area (560.6 $m^2 g^{-1}$) and nano-porous structure. However, as for the purpose of comparison, graphene was synthesized with a colloidal suspension of exfoliated graphene oxide sheets in water with hydrazine hydrate in various reaction times (12, 24 and 36 h). This method has obtained a six-layer graphene and graphene that was synthesized during 24 h reaction with hydrazine hydrate (GC24) had a higher quality in comparison with the other products. The GC24 had 195.97 $m^2 g^{-1}$ specific surface area and nano-porous structure. The as-synthesized graphene were then characterized by X-ray diffraction (XRD), Fourier transform infrared (FT-IR) spectroscopy, Raman spectroscopy, transmission electron microscopy (TEM), scanning electron microscopy (SEM) as well as BET measurements. The results demonstrated that this low-cost method for few-layer grapheme, e.g. three-layers, fabrication is reliable and promising.

1. Introduction

Carbon is a unique and very versatile element that is capable of forming different architectures with different properties and is capable of forming different architectures at the nano-scale. Over the last 40 years, new members of the carbon nanostructure family arose, and more are coming such as fullerene and carbon nanotubes [1]. The discovery of graphene is the last event of a long string of continuous advances in the science of carbon [2]. Carbon has four electrons in four hybridized bonding electrons ($2s^1 2p_x{}^1 2p_y{}^1 2p_z{}^1$) that are available to form covalent chemical bonds. In a two-dimensional carbon system for graphene, three carbon electrons from four hybridized bonding electrons and form strong in-plane sp^2 bonds consisting of a honeycomb structure, and a fourth electron spreads out over the top or bottom of the layer as a π electron [3].

Graphene has excellent in-plane mechanical, structural, thermal, and electrical properties. These properties make it interesting for many engineering applications. Graphene has been made by four different methods. The first is chemical vapor deposition (CVD) [4]. This method has started in 1970. The second is micromechanical exfoliation of graphite [5]. This approach that is also known as the Scotch tape or peel-off method followed on from earlier work on micromechanical exfoliation from patterned graphite. The third method is an epitaxial growth on electrically insulating surfaces such as SiC [6] and epitaxial growth on metal substrates [7] and the fourth method that recently is used is an intercalation component between graphite layers and then exfoliation of layers [8]. In this study the new method, the intercalation and exfoliation of graphite, is used. Graphite nanoplatelets have often been made from expanded graphite, which was produced from graphite intercalation compounds by rapid evaporation of the intercalate component at high temperatures. For instance, Cheng et al. has fabricated graphene by rapid thermal expansion of sulfuric acid-intercalated graphite [9].

In the graphite three atomic orbital s, p_x, and p_y of carbon on each carbon hybridize form strong covalent sp^2 bonds and the remaining p_z orbital on each carbon form a bond π with its neighboring carbons. The new route for synthesis graphene from graphite that is used in this study is an intercalation compounds method by oxidizing the graphite layers. The graphite layers loss its conjugation interlayer during the oxidization process and decorate their basal mostly with epoxies and hydroxyl groups, in addition carbonyl and carboxyl groups locate presumably at the edges (Lerf–Klinowski model) [10]. The graphite oxide obtains graphene

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oxide after sonication process. These oxygen functionalities add new hydrophilic properties to the graphene oxide layers and the water molecules can readily intercalate into the interlayer. Consequently, the water-intercalated graphite is obtained. The rapid heating of water-intercalated graphite results in its expansion and exfoliation due to the rapid evaporation of the intercalated water and evolution of gases produced by thermal pyrolysis of the oxygen-containing functional groups [11]. The expansion and exfoliation of water-intercalated graphite also can be resulted by eliminating the oxygen-containing functional groups via reducing agent [12].

However, the expansion and exfoliation of graphene oxide is a very complex phenomenon because of the multistep removal processes of intercalated H_2O molecules and oxygen-containing functional groups such as oxide groups of a carboxyl group, a hydroxyl group and epoxy group. Therefore, the expansion and exfoliation of graphene oxide and its results needs to be studied in detail.

In this study, at the first the graphene oxide is synthesized from graphite by modified Hummer's method [13]. The obtained graphene oxide retains a layered structure, it consists of oxidized graphene sheets, it is hydrophilic, and water molecules can readily intercalate into the interlayer space. The water-intercalated graphite is expanded and exfoliated using both thermal and chemical methods. The thermal method is used at different temperatures and the obtained graphene is characterized and compared in each temperature, then an optimum temperature is identified. The chemical method is also used in different times and the obtained graphene is characterized. A comparison between two chemically and thermally reduction methods of the graphene oxide and also the characteristics of the produced graphene by both methods, i.e. specific surface area, crystallinity, quality and number of layers are made herein. The product is characterized by Raman spectroscopy, scanning electron microscopy (SEM), X-ray photoelectron spectroscopy, transmission electron microscopy (TEM), Fourier transform infrared spectroscopy (FT-IR) as well as BET. The characteristics such as crystallinity, number of layers, porosity, volume of prose, diameter of pores, specific surface area are also measured. Finally the optimum condition for fabrication of a few layers, nanopores and high crystalline graphene with high surface area is identified.

2. Experimental

2.1. Preparation graphene oxide

The graphene oxide was prepared using a modified Hummers' method [13] as follow: First, 5 g of natural graphite (Merck), 3.75 g of $NaNO_3$ (99%, Sigma–Aldrich), and 310.5 g of H_2SO_4 (97%-Merck) were mixed in a 10-L Pyrex reactor with a water cooling system; this mixture was then stirred for 30 min. Next, 22.5 g of $KMnO_4$ (99%, Sigma–Aldrich) was carefully added to this mixture for 1 h. The resultant mixture was stirred for 5 days at the room temperature, after that 5 L of a H_2SO_4 aqueous solution (5%) was slowly added over a 1 h period. Then after stirring the solution for 2 h, 150 g of H_2O_2 (30%, Aldrich) was added to the mixture. Upon centrifugation, the graphite oxide slurry was obtained at the bottom and it was washed with a 3%-H_2SO_4/0.5%-H_2O_2 solution and then was rewashed with deionized water until the PH revived neutral. The graphite oxide was exfoliated in water by ultrasonication to produce a homogeneous graphene oxide water based suspension. Then the graphene oxide aqueous solution was filtered and dried in a vacuum oven at 70 °C for 24 h. The graphene oxide after the oxidation is a layered stack of wrinkled sheets, which completely exfoliates upon the addition of mechanical energy. This is due to the strength of interactions between water and the oxygen-containing (epoxide and hydroxyl) functionalities introduced into the basal plane during oxidation. The hydrophilicity leads water to readily intercalate between the sheets and disperse them. The water-intercalated graphene oxide which has obtained after above process is denoted as WI-GO in remaining of this paper.

2.2. Expansion and exfoliation WI-GO with thermally method

The WI-GO which is a brown color powder was placed into a quartz furnace with a crucible and treated thermally at different temperatures 200, 400, 600, 800 and 1000 °C for 30 min under a flow of laminar N_2 gas (99.999%). While the heating rate and the N_2 flow rate were 5 °C/min and 25 ml/min respectively. Graphene which were obtained at temperatures 200, 400, 600, 800 and 1000 were denoted as GT200, GT400, GT600, GT800 and GT100 receptivity.

2.3. Expansion and exfoliation WI-GO with chemically method

For exfoliation of WI-GO and to obtain graphene, the oxygen functional groups should be eliminated. To achieve this goal the reducing agents should be utilized. In this paper Hydrazin hydrate was used as reducing agent. It is the strongest reductant for graphene oxide while it has no reactivity with water. Then 100 mg of dried WI-GO was located in a 250 ml round bottom flask and 100 ml water was added, then it was sonicated by using ultrasonic bath cleaner symphony model for 10 min and 1 ml hydrazine hydrate was then added to the solution. The solution was heated in an oil bath and the solution was refluxed under magnetic stirring at 70 °C with different times 12 h, 24 h and 36 h. The product was filtered and washed with additional ethanol to remove residual hydrazine hydrate, and then it was dried in the oven at 100 °C for 3 h. Graphenes were obtained by the chemical method at different reaction times 12 h, 24 h and 36 were denoted as GC12, GC24 and GC36 respectively.

3. Characterization and measurements

The crystal structure studies of the solids were carried out by X-ray diffractions (Phillips PW 1840 X-ray diffractmeter with Cu-K radiation source). Graphene distance layer can be calculated based on Bragg's law [14,15]:

$$n\lambda = 2d_{(h\ k\,l)} \sin(\theta) \quad (1)$$

where λ is the wavelength of the X-ray, θ is the scattering angle, n is an integer representing the order of the diffraction peak, d is the interplane distance of the lattices, and $(h\,k\,l)$ are Miller indices. The mean crystallite size of the powder composed of relatively perfect crystalline particles can be determined via so-called Scherrer equation [14,15]:

$$L_{\mathrm{h\ k\,l}} = \frac{k\lambda}{\beta_0 \cos\theta} \quad (2)$$

where $L_{h\,k\,l}$ is the mean dimension of the crystallite perpendicular to the plane $(h\,k\,l)$; β_0 is the integral full widths at half maximum in radians; K is a constant dependent on the crystallite shape (0.89). However, the number of graphene layers (N) can be obtained using the following equation [16]:

$$N = \frac{L_{h\ k\,l}}{d_{h\,k\,l}} \quad (3)$$

Furthermore, the nitrogen adsorption/desorption isotherms were recorded by using Belsorp miniII (Bel, Japon) at 77 K. Prior to physisorption measurements, the samples were out gassed at 373 K for 20 h. The specific surface area and pore volume were calculated by using the BET and BJH methods.

The morphology of the products was examined by transmission electron microscopy (CM30-Philip) and scanning electron microscopy (S4160-Hitachi Japan). The Samples were prepared for TEM by dropping the products on a carbon coated copper grid after ultrasonic dispersing in absolute ethanol and allowed them to dry in the air before analysis.

The Raman spectroscopy stands as a powerful characterization technique of all types of carbon nanostructures. The most prominent features in the Raman spectra of sp^2 hybridized carbon materials are the G-band appearing at ca.1580 cm^{-1} and the 2D-band at 2400–2600 cm^{-1}. When carbon materials have a certain amount of disorder or edges within the structure another band appears at 1200–1400 cm^{-1}, the so-called D-band [17,18]. The Raman spectroscopy indicates the number of graphene layers via a change in the position and intensity of the G and the 2D band [19]. The Raman spectra were performed on a Senterra model of Bruker Company (Germany) and a 785 nm laser source was used a Raman spectrometer.

The method that is used for characterizing material is Fourier transform infrared (FT-IR). In infrared spectroscopy, IR radiations are passed through a sample. Some of the infrared radiations are absorbed by the sample and some of them are passed through (transmitted). The resulting spectrum represents the molecular absorption and transmission, creating a molecular fingerprint of the sample. The Fourier transform infrared (FT-IR) spectra were recorded in this study on a Bruker IFS 88 Fourier Transform Infrared spectrophotometer with KBr pellets in the 4000–400 cm^{-1} region.

4. Result and conclusion

4.1. Thermally reduction

The graphene oxide is thermally unstable and electrically insulating and thus cannot be used, without further processing, as a conductive nanomaterial. Notably, it has been demonstrated that the thermal stability and electrical conductivity of the graphene oxide can be restored close to the level of graphite by reduction. In this work, a heat treatment was used for expansion and exfoliation water intercalated graphene oxide to obtain graphene nanosheet. The graphene oxide starts to lose mass upon heating and the major mass loss occurs at 200 °C, presumably due to pyrolysis of the oxygen-containing functional groups, yielding CO, CO_2, and steam [11,20]. Hence, the thermal decomposition of graphene oxide can be accompanied by a vigorous release of gas, resulting in a rapid thermal expansion of the material. This is evidenced by both a large volume expansion and a larger mass loss during a more rapid heating regime. No significant mass loss is detected when this material is heated up to 200 °C.

To investigate the oxidation and reduction processes, a XRD pattern was recorded from WI-GO and the synthesized graphene by thermally method on Fig. 1. The structural parameters such as: 2θ plane (0 0 2), Interlayer distance (d) is calculated by Bargg's low (Eq. (1)), full widths at half maximum (FWHM), crystal size that is calculated by Eq. (2) and number of graphene layers is calculated by Eq. (3) are estimated from XRD pattern and they are summarized in the Table 1. The XRD pattern of WI-GO shows a strong and sharp peak at $2\theta = 11.68°$ corresponds to an interlayer distance of 7.61 Å. The graphene oxide has a larger interlayer distance than graphite (d graphite ~3.4) because of H_2O molecules and various oxide groups intercalate between its layers. The interlayer distance of the graphene oxide is in range of ~5–9 Å, depending on the number of intercalated water molecules [3]. The XRD patterns of graphene was recorded via expansion and exfoliation WI-GO

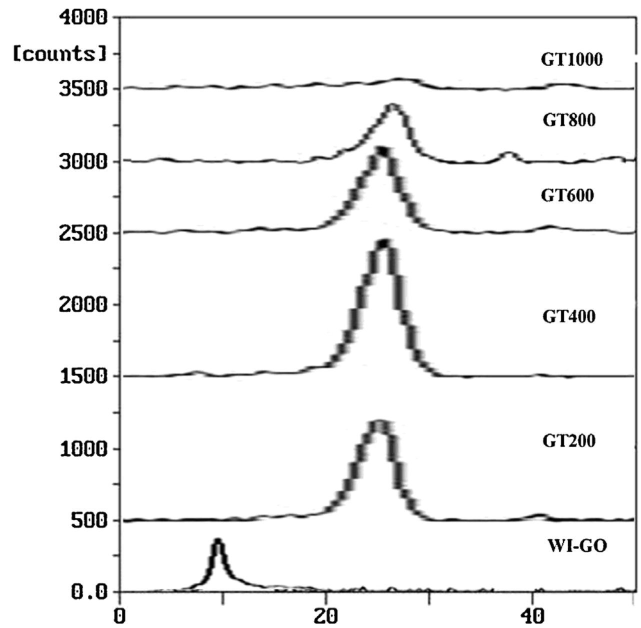

Fig. 1. X-ray patterns of WI-GO, GT200, GT400, GT600, GT800 and GT1000.

at temperatures 200, 400, 600, 800 and 1000 °C (GT200, GT400, GT600, GT800, GT1000) has a strong peak at $2\theta = 26–27°$. Originally, the XRD pattern of graphene has a strong peak at $2\theta = 26–27°$ [16,21]. The GT200, GT400, GT600, GT800, GT1000 all are graphene according to the XRD patterns. The drastic changes in interlayer distance of 7.61–3.43 Å (Table 1), for the values of WI-Go and GT200 is due to the exfoliation of GO sheets after rapid vaporization of the intercalated water molecules. According to the values of Table 1, the interlayer distance is the same in GT200–GT400 but the interlayer distance of GT400, GT600, GT800 and GT1000 descend respectively 3.43, 3.41, 3.40 and 3.38 Å. The water molecules are removed at temperatures below 200 °C and most of the hydroxyl and the carboxyl groups are removed from WI-GO at temperatures between 500 and 600 and a few hydroxyl and epoxide groups which still found as residues are removed gradually at temperatures between 600 and 1000 °C [16]. However, the interlayer distance of GT800 (3.40 Å) is the closest to that of conventional graphene (3.4 Å) [22]. In the Table 1, we can see the drastic increasing in FWHM of WI-GO and GTs that is resulting from folding and wrinkling of the graphene sheets. The FWHM decreases as the temperature increase from 200 to 800 °C and then increases as temperature increase from 800 to 1000 °C. Therefore, the quality of graphene sheets improves as the temperature increase from 200 to 800 °C and then impairs as temperature increase from 800 to 1000 °C.

The FT-IR spectra of the graphite, WI-GO and GTs are shown in Fig. 2. The IR spectrum of the starting graphite is detected only one peak at 1565 cm^{-1} attributable to C=C (graphitic carbon atom vibrations). Several peaks of WI-GO can be observed in Fig. 4: the broad and intense absorption at 3405 cm^{-1} originated from O—H

Table 1
X-ray structural parameters of WI-GO, GT200, GT400, GT600, GT800 and GT1000.

Samples	2θ	FWHM (°)	Layer distance (Å)	Crystal thickness (nm)	Number of layer≈
WI-GO	11.68	3.0	7.61	2.63 ± 0.02	3
GT200	26.68	9.4	3.43	0.86 ± 0.02	3
GT400	26.68	9.3	3.43	0.87 ± 0.02	3
GT600	26.82	8.5	3.41	0.95 ± 0.02	3
GT800	26.97	8.0	3.40	1.01 ± 0.02	3
GT1000	27.12	8.7	3.38	0.93 ± 0.02	3

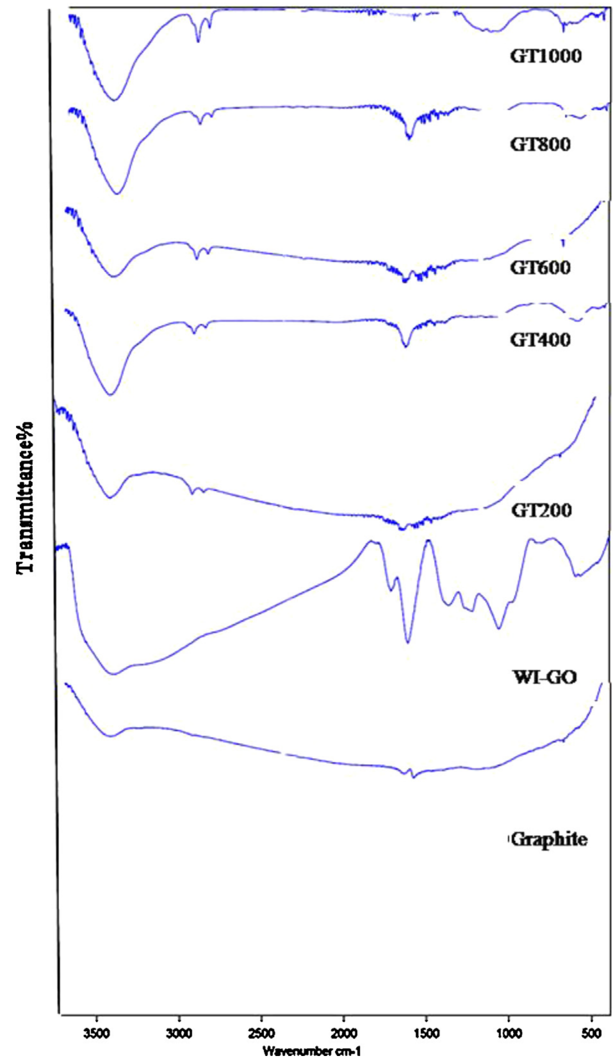

Fig. 2. FT-IR spectra of graphite, WI-GO, GT200, GT400, GT600, GT800 and GT1000.

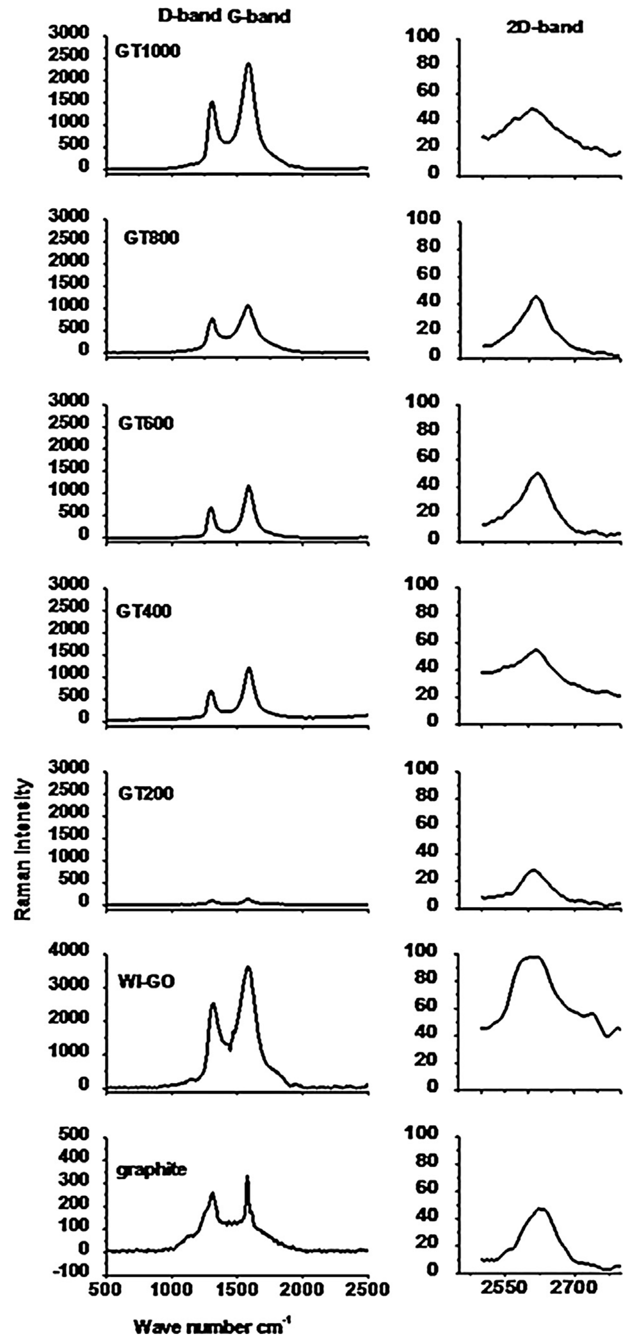

Fig. 3. Raman spectra of graphite, WI-GO, GT200, GT400, GT600, GT800 and GT1000.

stretching vibration, a peak at 1702 cm^{-1} due to the C═O stretching of carboxylic groups situated at the edges of GO sheets. The stretching vibration peaks of C—O (epoxy) and C—O (alkoxy) are observed at 1410, 1216, and 1040 cm^{-1}, respectively. The peak at 1585 cm^{-1} can be attributed to in-plane C═C bonds and the skeletal vibration of the graphene sheets that confirming the successful oxidation of graphite [3,27]. The FTIR of expansion and exfoliation of water-intercalated graphite by heating them at different temperatures indicates that the heating at 200 °C removes the most carboxyl groups, heating at 400, 600 and 800 °C removes the residual carboxyl and partial hydroxyl groups, heating at 800 °C removes the residual hydroxyl groups and the epoxy group and heating at 1000 °C cracking of aromatic C═C bonds.

The Raman spectroscopy of graphite, WI-GO, GT200, GT400, GT600, GT800 and GT1000 is shown on Fig. 3. The Raman spectrum of the pristine graphite displays a G peak at 1580 cm^{-1}. These bands are present in all poly aromatic hydrocarbons. The G peak is due to the bond stretching of all pairs of sp^2 atoms in both rings and chains [2]. The Graphite's Raman spectrum displays also a weak D peak at 1312 cm^{-1} and a 2D peak at 2623.5 cm^{-1}. In the Raman spectrum of WI-GO, the G band is broadened and shifted to 1586.5 cm^{-1}. In addition, the D band at 1318 cm^{-1} becomes prominent, indicating the reduction in size of the in-plane sp^2 domains, possibly due to the extensive oxidation. The 2D peak observed at 2603.5 cm^{-1} from WI-GO also shows a significant change in the shape compared to graphite.

The Raman spectrum of the GT200, GT400, GT600, GT800 and GT1000 also contains both G and D bands. The G band of GTs downshift to the position close to the G band of graphite and this is attributed to graphitic self-healing similar to what was observed from the sharpening of the G peak and the intensity decrease of the D peak in heat-treated graphite [23,24]. Yoon and et al. explained that the G band frequency of single-layer graphene is 1585.5 ± 1 cm^{-1}, and that of bi-layer graphene is 1581.5 ± 0.5 cm^{-1}. The frequency for other thicknesses is 1582.5 ± 1 cm^{-1} [18,25]. The result of Table 2 prove that the G band frequency is ca. 1582.5 ± 1 cm^{-1} and that is why the number of GTs layer is more than two. According to G band frequency and X-ray results the GTs have three layers. The G band intensity is very low at GP200

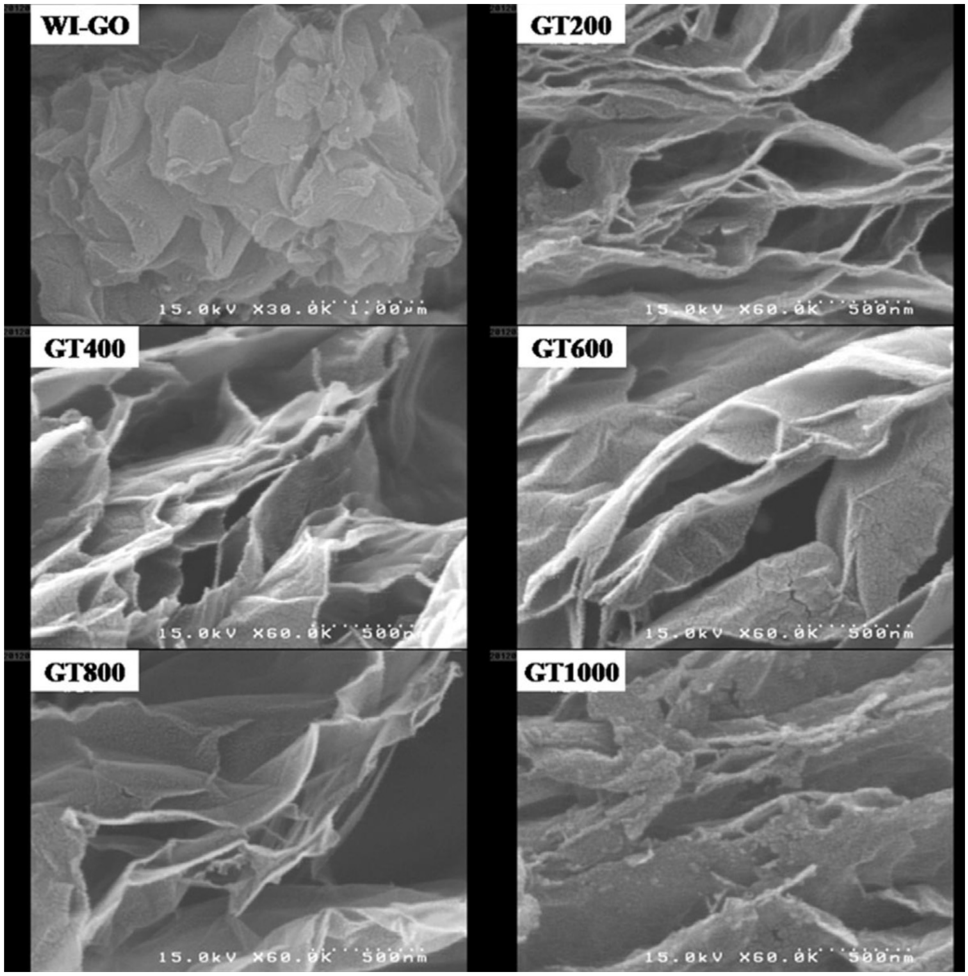

Fig. 4. FE-SEM image of WI-GO, GT200, GT400, GT600, GT800 and GT1000.

but GT400, GT600, GT800 and GT1000 has a prominent G band while their G band intensity approximately are so close together. Additionally, the order/disorder carbon ratio (I_D/I_G) of GT200 has increased comparing to the WI-GO (Table 2). This change suggests a decrease in the average size of the sp^2 domains upon heat treatment of WI-GO [24]. On the other side I_D/I_G intensity ratio decreases with increase in temperature from 200 to 800 °C and then decreases with increase in temperature to 1000 °C which implies that the disordering portion of the two-dimensional carbon backbone is increases in this temperature range because of cracking C=C in-plan graphite sheet that it is demonstrated via the FTIR of GT1000.

Originally, the Raman spectrum of graphene has another peak at ca. 2600 cm^{-1} that it is called 2D band and can be used as an exact fingerprint for the number of graphene layers. The 2D band can be explained by a double resonance Raman process and has a close correlation with the electronic band structure of the graphitic materials [2,17,25]. For single layer graphene, the 2D band can be fitted with a sharp and symmetric peak while the 2D band shifts to higher wave number values and becomes broader when the graphene thickness increases from single to multi layer graphene. Unfortunately, the differences in the 2D band between two and a few layers of graphene sheets are not unambiguous in the Raman spectra [2,17].

Fig. 4 is shown Fe-SEM of WI-GO, GT200, GT400, GT600, GT800 and GT1000. During the expansion and exfoliation graphene oxide by heating up, WI-GO sheets are individually exfoliated at temperatures around 200 °C due to vaporization of intercalated water molecules. Since the exfoliated WI-GO sheets are thermodynamically unstable, they couple with one another to stabilize into thick layers that yield micro or macro wrinkling and folding. The GTs have macropores like crater surrounding, the macropores and folded sheets owing to the vaporization of intercalated H_2O molecules, as

Table 2
The G band and D band position and their intensity, intensity of D band G band ratio and 2D band positions of graphite, WI-GO, GT200, GT400, GT600, GT800 and GT1000.

Samples	D-band position (cm^{-1})	G-band position (cm^{-1})	I_G	I_D/I_G	2D-band position (cm^{-1})
Graphite	1312.0	1580.0	331.08	0.77	2623.5
WI-GO	1318.0	1586.5	3620.06	0.69	2603.5
GT200	1317.5	1582.0	133.85	0.71	2611.0
GT400	1310.0	1583.0	1198.83	0.71	2619.0
GT600	1300.5	1583.5	1171.79	0.69	2620.0
GT800	1298.5	1582.5	1083.36	0.62	2614.5
GT1000	1309.0	1584.0	2397.49	0.93	2608.0

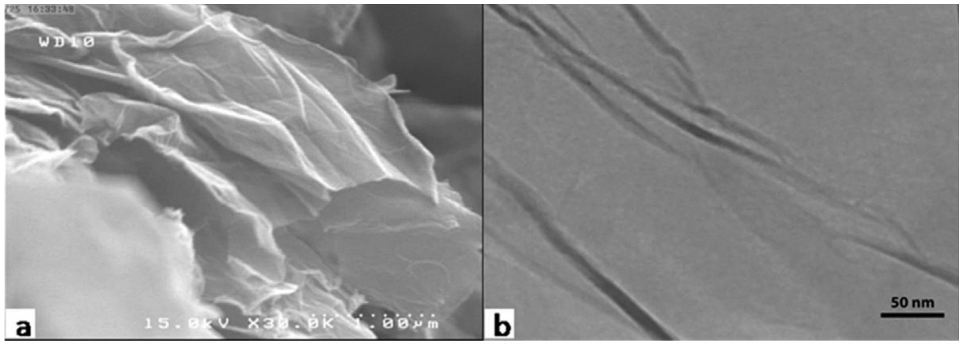

Fig. 5. (a) FE-SEM and (b) high resolution TEM of GT800.

shown in Fig. 4. The macro folded GT200 is partially unfolded during the increasing temperature and largely unfolded above 1000 °C but the basal is destroyed. The unfolding process of GTs with increasing temperature may be attributed to breaking the conjunction of inter-layer and bonds between functional groups and the basal owing to the effects of thermal annealing and the removal of oxygen functional groups. However, the basal destruction at 1000 °C owing to that the residual hydroxyl groups need a carbon source to yield CO gas and the carbon comes from an ordered aromatic C=C structure. Therefore, the I_D/I_G intensity ratio will be increased as shown in the Raman spectra.

4.1.1. Temperature optimization

According to the result of analysis that is explained before, the water molecules intercalated between graphite layers and oxygen functional groups were located on the basal resulted a high quality graphene oxide. Then via heat treatment of water intercalated graphene oxide, graphene was also obtained. The qualities of the GTs improved and the disorder decreased as the temperature increasing from 200 to 800 °C. The increasing temperature until 1000 °C causes to break in-plane carbon bond and the qualities of GT1000 is decreased and the graphene sheets were destroyed. Graphene was obtained via the exfoliation and expansion WI-GO by thermal method has three layers. The graphene oxide is not exfoliated any more with temperatures above 200 °C, but the higher temperature removes the oxide groups. Consequently, the optimum temperature for exfoliation and expansion graphene oxide to obtain a high quality tri-layer graphene is 800 °C. The GT800 has some folding and wrinkling areas as shown in the FE-SEM and high-resolution TEM images (Fig. 5) but it has high-quality structures based on the XRD and the Raman analysis. Furthermore, the GT800 has the flat area between the micro-folding areas and shows the in-plane lattice ordering of a two-dimensional carbon network, and has a large size flake as shown in the TEM.

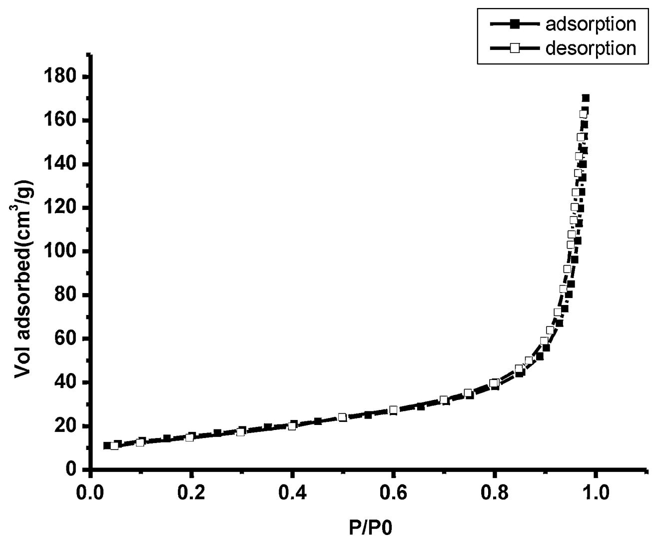

Fig. 6. Low-temperature adsorption–desorption isotherms of nitrogen for GT800.

The N_2 adsorption–desorption isotherms of graphene sheets are shown in Fig. 6. The Brunauer–Emmett–Teller (BET) specific surface area of graphene sheets (GT800) is about 560.6 $m^2\,g^{-1}$. However, it is still lower than the theoretical specific surface area for completely exfoliated and isolated graphene sheets (~2620 $m^2\,g^{-1}$ [20]), due to the agglomeration, overlapping and fusing the graphene sheets can result lower surface area. The isotherm observed in the Fig. 6 can be considered as typical type III isotherms [26]. This isotherm is the characteristic of weak gas–solid interactions. This weakness

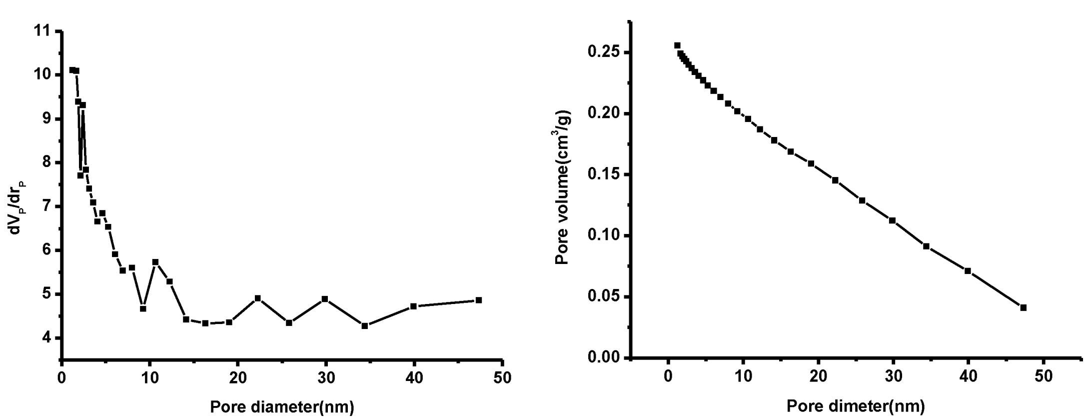

Fig. 7. Pore volume and pore diameter distribution of GT800.

Table 3
X-ray structural parameters of GC12, GC24 and GC36.

Samples	2θ	FWHM (°)	Layer distance (Å)	Crystal thicness (nm)	Number of layer≈
GC12	26.08	4.1	3.50	1.97	6
GC24	26.94	3.7	3.40	2.18	6
GC36	27.53	3.9	3.33	2.07	6

causes the uptake at low relative pressures to be small; but once a molecule has become adsorbed, the adsorbate–adsorbate forces will promote the adsorption of further molecules, and the resulting isotherms will become convex to the pressure axis. The BJH pore size distribution is shown in Fig. 7. The main pore size belongs to mesoporous (2–50 nm) that mesoporous materials are an important category of nanoporous materials. Furthermore, the total pore volume of graphene sheets is 0.263 cm^3/g at $P/P_0 = 0.98$ and the average pore diameter is about 18.76 nm calculated by the BET model.

4.2. Chemically reduction

An important property of graphene oxide, brought out by the hydrophilic nature of the oxygenated graphene layers, is its easy exfoliation in aqueous media. As a result, graphene oxide readily forms stable colloidal suspensions of thin sheets in water [28]. In this section, the authors used a chemical treatment also for expansion and exfoliation aqueous graphene oxide solution to obtain graphene nanosheet. For this purpose, a variety of reducing agent may be used for expansion and exfoliation aqueous graphene oxide solution. The hydrazine monohydrate has no reactivity with water while the other reductant do and this makes it an attractive option for reducing aqueous dispersion of graphene oxide [20].

Fig. 8 shows XRD pattern and Table 3 shows XRD parameters of synthesized graphene by chemical method. As explained before the XRD pattern of WI-GO shows a strong and sharp peak at $2\theta = 11.68°$ correspond to an interlayer distance 7.61 Å. However, The XRD

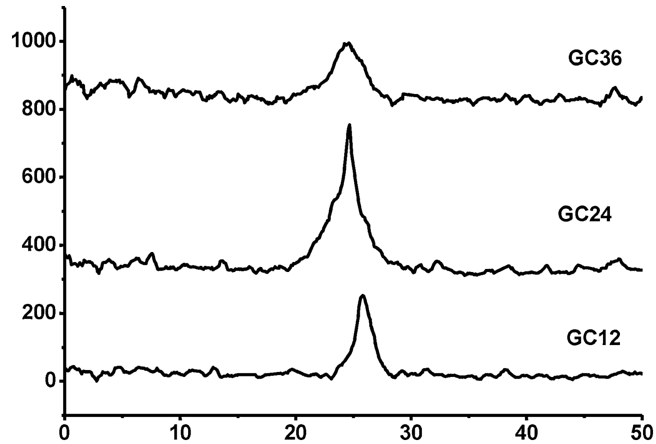

Fig. 8. X-ray patterns of GC12, GC24 and GC36.

patterns of graphene was obtained via expansion and exfoliation WI-GO via reaction by Hydrazine hydrate at different reaction times 12 h, 24 h and 36 h (GC12, GC24, and GC36) has a strong peak at $2\theta = 26–27°$ (Table 3). The drastic changes in interlayer distance of 7.61–3.50 Å after 12 h reaction for the values of WI-Go and GT200 is due to the exfoliation of WI-GO and removing the intercalated water molecules. When the reaction was taken for 24 h the interlayer distance of GC24 (3.40 Å) is the closest to that of conventional graphene (3.4 Å) [22]. The FWHM in XRD pattern of GC24 is the smallest one. Therefore, the quality of obtained graphene after 24 h reaction is the best and according to the XRD pattern a six-layer graphene is obtained.

The FT-IR spectra of the GC12, GC24 and GC36 are shown in Fig. 9. The IR spectrum of the GC12, GC24 is detected a peak at 1570 cm^{-1} attributable to C=C (graphitic carbon atom vibrations) but this peak is weak in the IR spectra of GC36 that is implied aromatic C=C bonds is cracking due to long time reaction. The IR spectra of GC12, GC24 and GC36 also have a peak at 1214 cm^{-1} due

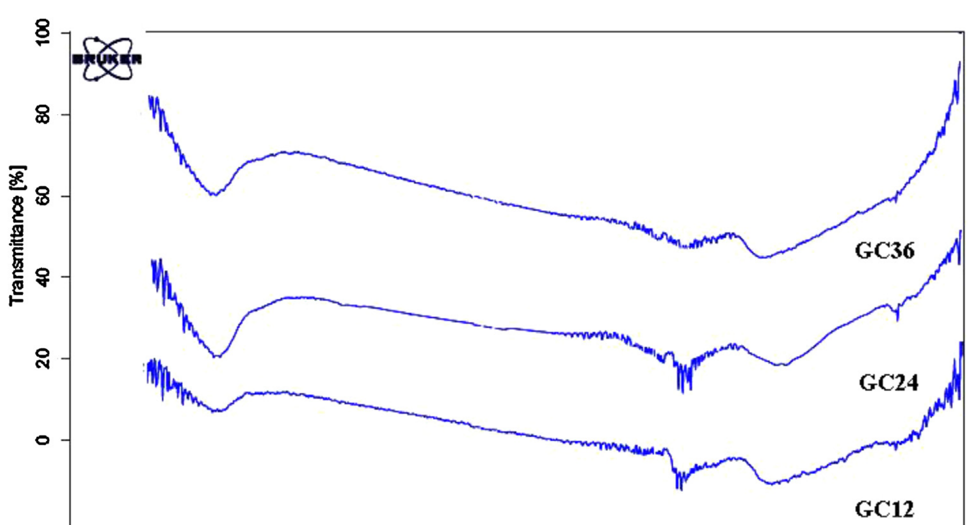

Fig. 9. FT-IR spectra of graphite, GC12, GC24 and GC36.

Table 4
The G band and D band position and their intensity, intensity of D band G band ratio of GC12, GC24 and GC36.

Samples	D-band position (cm^{-1})	G-band position (cm^{-1})	I_G	I_D/I_G
GC12	1310.0	1584.0	188.50	0.67
GC24	1305.3	1581.6	505.51	0.61
GC36	1300.1	1580.1	683.27	0.68

to the stretching C—N bond. The nitrogen tends to remain covalently bonded to the surface of graphene oxide, likely in the form of hydrazones, amines, aziridines or other similar structures (Fig. 10).

The Raman spectrum of the GC12, GC24 and GC36 has both G and D bands. The G band of GCs down-shift to the position close to the G band of graphite and this is attributed to graphitic self-healing similar to GTs [23,24]. The G band intensity is very low at GC12 but GC24 and GC36 has a prominent G band because the duration of 12 h is not enough long for a reaction. Additionally, the I_D/I_G ratio of GC24 is the smallest comparing to GC12 and GC36 (Table 4). Consequently, the quality of GC24 is better than GC12 and GC36 that has been demonstrated by XRD and FTIR spectra.

Fig. 11 illustrates FE-SEM of GC12, GC24 and GC36, the significant folding or overlapping of platelets can be observed. The WI-GO sheets are not completely exfoliated during of 12 h reaction by hydrazine as show the FE-SEM of GC12 (Fig. 11), while WI-GO sheets are more exfoliated for GC24 and the edge of sheets are destroyed for GC36 as seen on FE-SEM. This was already confirmed by previously discussed analysis.

4.2.1. Optimization of reaction time

Based on the Lerf–Klinowski model the graphene oxide contains of the most of the oxygen functionalities that should be present in the form of either hydroxyl or epoxide groups. Additionally, the graphene oxide contains a number of carbonyl-containing oxygen functionalities such as lactones and anhydrides [10,28]. The Hydrazine can react with anhydrides and lactones to form

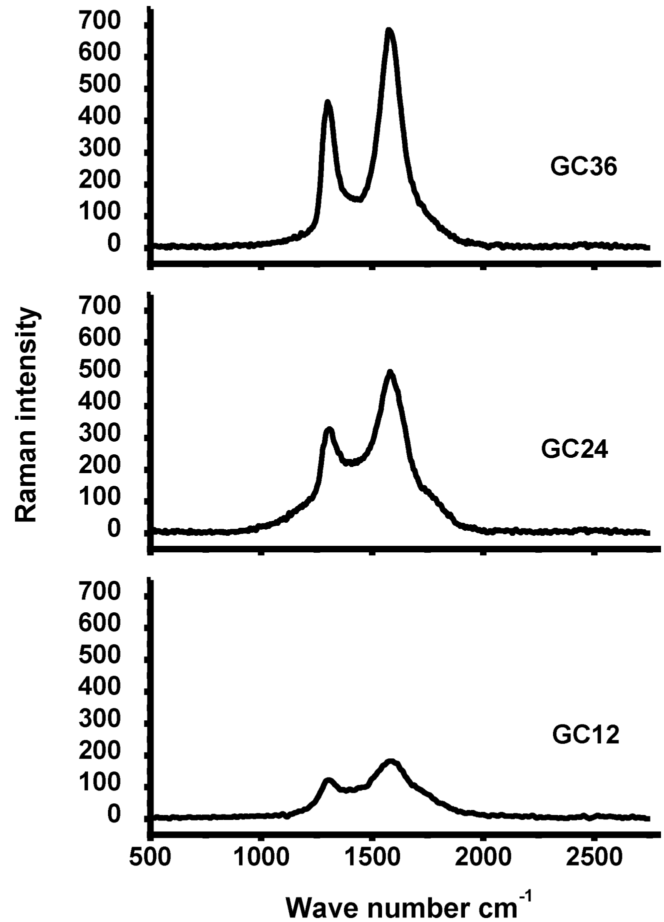

Fig. 10. Raman spectra of GC12, GC24 and GC36.

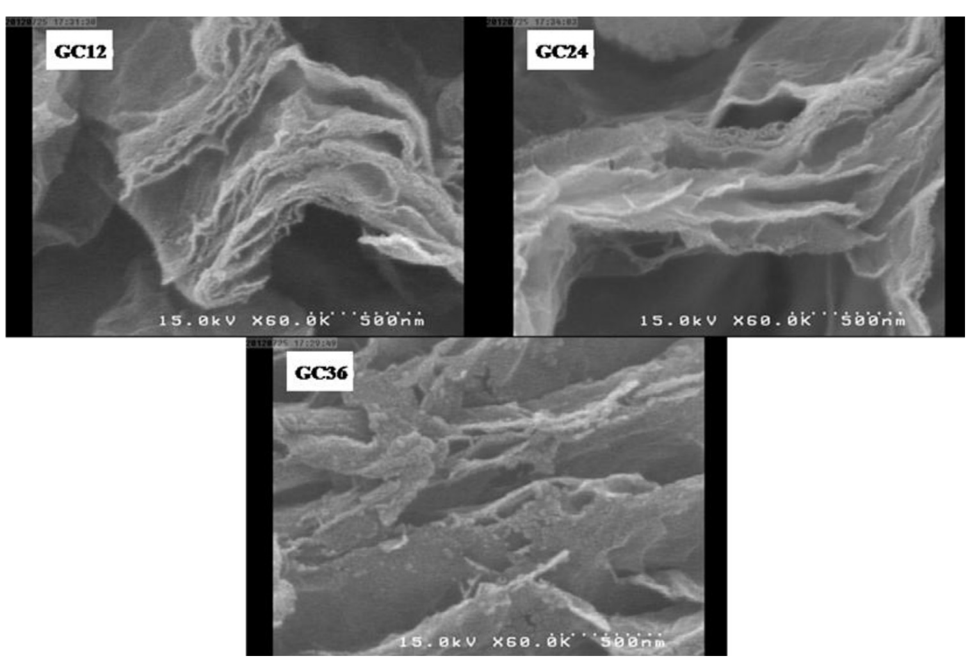

Fig. 11. FE-SEM image of GC12, GC24 and GC36.

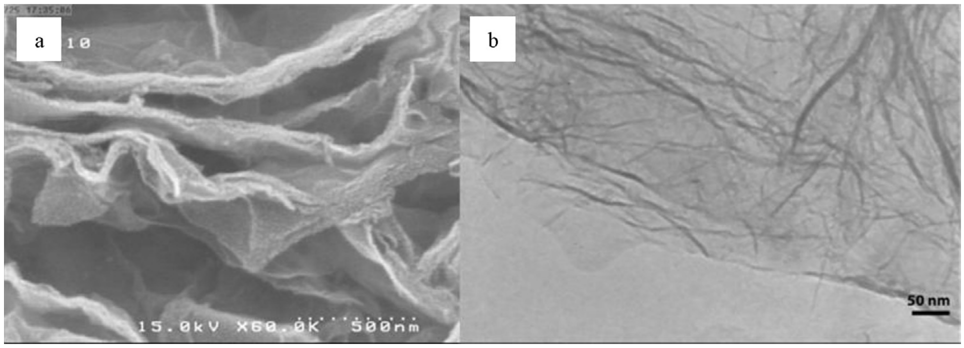

Fig. 12. (a) FE-SEM and (b) high resolution TEM of GC24.

hydrazides to yield a deoxygenated sp^2-carbon [11,20]. According to the analysis that were explained, the most of functional group of graphene oxide is removed during 12 h reaction with hydrazine and the reaction during the 24 h causes improvement in the quality of graphene sheets but 36 h reaction impair the quality of graphene sheets. Furthermore, one of the disadvantages of using chemical methods (hydrazine) is nitrogen tendency to remain covalently bonded to the surface of graphene oxide. Residual C—N groups have a profound effect on the properties of the resulting graphene. These impurities are increasing by extending the time of reaction. Therefore, the optimum time for expansion and exfoliation water intercalated graphene oxide to obtain graphene is 24 h. According to the Table 3 the synthesized graphene by chemical method is a six-layer graphene.

Fig. 12 illustrates the FE-SEM and high-resolution TEM images of GC24. The GC24 has micro and macro wrinkling and folding at the edge of sheets while the plates have a large size.

To investigate the surface area and porosity of graphene the N_2 adsorption–desorption isotherms of GC24 are shown in Fig. 13. The Brunauer–Emmett–Teller (BET) specific surface area of GC24 is about 195.97 m^2 g^{-1} is lower than the theoretical specific surface due to existing wrinkling, folding and crumpled sheets. The isotherms observed in this figure can be considered as typical type III isotherms [26]. It has a hysteresis loop at a relative pressure which indicates that the graphene sheets are porous and none overlapping of the adsorption and desorption branches. According to Barret–Joyner–Halenda (BJH) results (Fig. 14), the main pore size belongs to nanopores (2–50 nm) and the total pore volume is 0.0651 cm^3/g and the average pore diameter is about 13.28 nm at $P/P_0 = 0.98$.

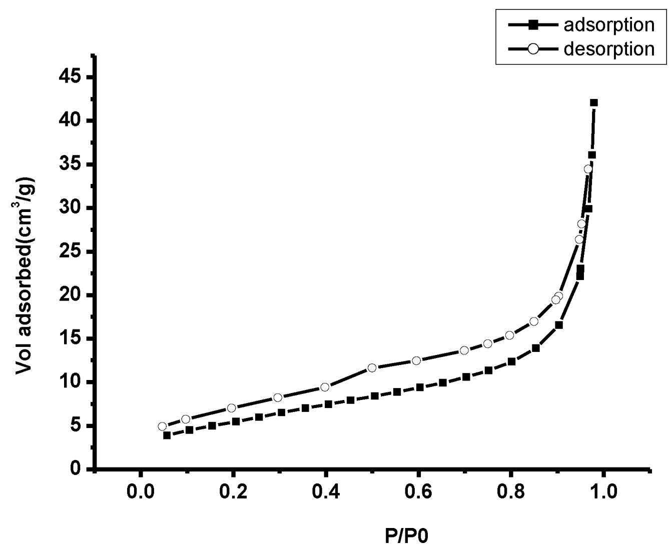

Fig. 13. Low-temperature adsorption–desorption isotherms of nitrogen for GC24.

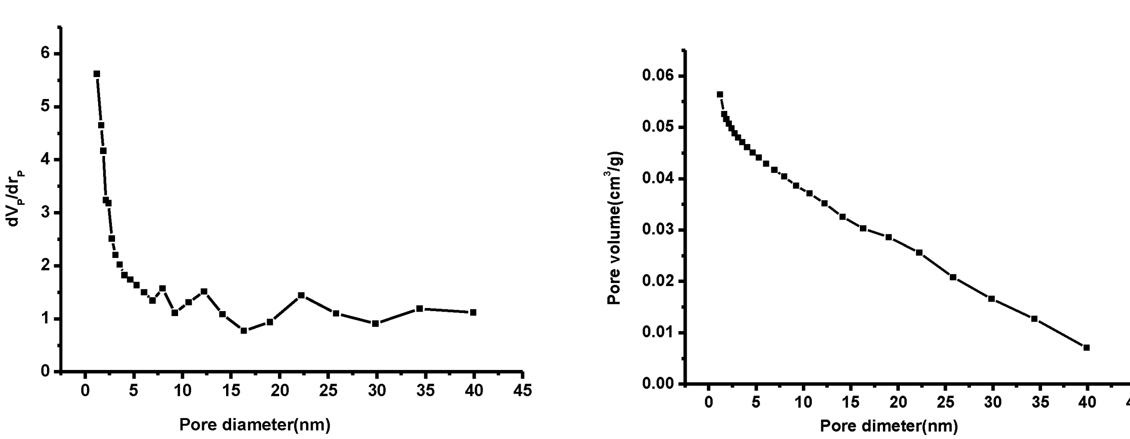

Fig. 14. Pore volume and pore diameter distribution of GC24.

5. Conclusions

Graphene has excellent properties which make it interesting for many engineering applications [29,30]. The method of utilizing water intercalated graphene oxide for synthesizing graphene is a low-cost and massive scalability, while the starting material is just simple graphite and the technique can easily be scaled-up. In this method at the first the graphene oxide was produced then it has to be reduced and exfoliated to regular graphene. In this study both thermal and chemical methods were used. According to the analysis of GT800 and GC24, they had a similar disorder/order carbon (I_D/I_G). Therefore a high quality graphene was produced by both methods. While graphene that was synthesized by thermal method had three layers, another one had six layers. It is also clear that the properties of graphene depend on the number of graphene layers. According to the discussion of BET analysis, the specific surface area of graphene that was synthesized by thermal method was more than another one owing to the number of layers. However, graphene synthesized by both methods had nano-porous structure. Nano-porous materials have numerous applications in catalysis, separation and many other fields. The synthesis of these materials is of considerable interest and continually will be subject of our further publication as well.

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